

E-Portfolio and reflection: Machine Learning Module

Submitted by:

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GitHub: <https://msalhamdani99-svg.github.io/e-portfolio/Machine-Learning.html>

E-portfolio

1. Introduction

This e-portfolio is intended to show my learning experience in the process of taking the Machine Learning module, my personal contributions, learning technical skills, and my reflections on the difficulties that I faced in the process of applying machine learning algorithms to the real-life context. Regression models, clustering, and ethical concerns of AI technologies were some of the topics discussed in the module. I was working on the analysis of the prices of Airbnb listings with different machine learning models and the prospects and risks of AI writers.

This e-portfolio will be a compilation of the work that I have done, as well as a critical reflection on my experiences in the course of the module. It also includes a reflective section, which entails my personal development, learning outcomes, and reflections of the module, as well as the reflection of the technical work I performed, including preprocessing of the dataset, the choice of a model, and its evaluation. Moreover, peer and tutor feedback is also provided in this e-portfolio and has helped me to learn.

2. Project Overview and Key Contributions

Airbnb Price Prediction

The main project consisted of predicting the prices of Airbnb listings in New York City (using machine learning). The data was used to build regression models, and it consisted of various attributes like room type, neighbourhood group, and availability, among others.

Key Steps:

Data Preprocessing: Data cleaning was done by taking care of missing data and skewed data (log transformation of the price variable).

Model Selection and Evaluation: I have compared three regression models, namely Linear Regression, Ridge Regression, and Random Forest Regression. I found out that the largest model in price prediction was the Random Forest since it controlled the non-linear associations.

Model Evaluation: I have cross-validated and hyper-parameterised the models and used the measures of R^2 and Mean Squared Error (MSE) to evaluate the models.

The most important lesson learned during the project was that it is necessary to choose the appropriate model, and it is imperative to analyse its performance properly. The effectiveness of the Random Forest model in the task of complex relationships proved the effectiveness of a non-linear algorithm in real-life.

AI Writers: Risks and Benefits

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Key Points:

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- **AI in Administrative Work:** AI can also be implemented to automate administrative work, which usually presupposes the creation of reports, emails, and regular documents, with the assistance of AI-based language models. This is able to improve efficiency and scalability (Bhat, 2023).

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- **AI in Creative Writing:** AI can be used to help with creative dumps and text generation; however, it is feared that it will destroy creativity, undermine authorship, and generate biased or inaccurate writing (Begum, 2025).
- **Ethical Consequences:** Artificial intelligence in writing can be inaccurate or biased, and that can prove to be catastrophic in fields such as healthcare, law, and policy development. Such threats can be mitigated only with the help of human control (Ganai and Naikoo, 2025).

The debate intensified the necessity to apply AI in a sensible manner, especially in areas where precision and creativity are significant. Other ethical concerns that I took into consideration were prejudice and the influence that AI can have on intellectual property.

3. Technical Evidence and Skills Development

Artefact 1: Random Forest Regression Implementation

The application of the Random Forest model to the Airbnb data set presented a good demonstration of my skills in the application of machine learning algorithms. Pre-processing data, model optimisation, and performance analysis of the model gave me a chance to demonstrate that I am aware of the necessity to apply machine learning to predictive tasks.

Artefact 2: K-Means Clustering

I also applied K-Means clustering to divide the listings according to different characteristics, in addition to regression. The clustering analysis helped in deriving four market segments of Airbnb listings. This was a good learning experience on the way hosts could maximise their pricing strategy depending on the location, the type of room and other aspects.

Artefact 3: AI Writer Ethical Analysis

I was able to evaluate the overall implications of AI technologies in society critically. I have touched upon concerns like bias in AI outputs, the danger of hallucinations and how ethical guidelines should be applied to the development and implementation of AI.

4. Feedback and Collaboration

During the module, I got feedbacks with peers as well as tutors. This feedback helped me to improve my perspective on machine learning methods and the ethical implications of AI technologies.

- **Peer Feedback:** My peers gave me valuable feedback on model evaluation methods and helped me to debug problems when implementing the Random Forest regression and K-Means clustering.
- **Tutor Feedback:** The information provided by my tutor about the ethical issues of AI writers and the necessity of human control influenced my perception of the drawbacks of AI and the significance of transparency.

Cooperation with colleagues also allowed me to understand how to cooperate in a team, combining different points of view and approaches to address complicated issues.

5. Overall analysis:

During the duration of the Machine Learning module, I have shown significant improvement in both technical and teamwork abilities. The module involved the implementation of machine learning methods to practical issues, such as estimating the price of Airbnb listings, data segmentation through the clustering technique, and the possibilities and dangers of applying AI

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writers. These artefacts can be used as confirmation that I have used machine learning algorithms, including Random Forest Regression, Ridge Regression, and K-Means clustering. Working on the Airbnb price prediction project was an important learning experience. It involved improving my understanding of regression models and their use in complex data. I could achieve a substantial improvement in the predictive performance of the model by choosing a suitable model, using log transformation, and adjusting the hyperparameters. The other notable part of the module was the ethical discourse of AI writers. I investigated the social consequences of machine-generated content, especially the possibility of partiality and misinformation. This led to a sensitive understanding of the ethical concerns of machine learning technologies and their application in life, particularly in such sensitive fields as medicine, law and media. Through these activities, I gained technical skills in machine learning algorithms within the larger context of their implications on society.

As well, I was engaged in team discussions and provided some information about data preprocessing, model evaluation, and ethical aspects. I also worked with my colleagues and cooperated on various aspects, including data cleaning, feature extraction and clustering analysis. Peer and tutor feedback played an instrumental role in assisting me to master my style. The two sources helped me with the troubleshooting of my model implementation, analysis, and understanding of my work better by providing positive feedback. It is due to this process that the significance of feedback and the learning experience were brought into the limelight, and I was challenged to approach problems differently.

Comparing the final project (Unit 11) to the first proposal (Unit 6), I understood that my knowledge about machine learning had changed considerably. When I was starting the module, I was largely referring to the theoretical aspect of algorithms, but towards the end of the course, I got a clearer

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picture of the reality behind the usage of such approaches. My knowledge of model selection, performance assessment, and hyperparameter optimisation has increased, and a more sophisticated approach to the final project has been developed. The other competency that I acquired was the ability to balance between technical analysis and ethical concerns by ensuring that I not only utilise the machine learning techniques effectively, but also pay close attention to the impact of the techniques on society.

On the whole, this module has affected my professional and personal growth significantly. It has already equipped me with a solid background in machine learning and enabled me to critically reflect on the ethical and social implications of AI. Now I am more certain about my further application of these competencies and my skills to take part in the discussion of the responsible development and implementation of AI technologies.

6. Conclusion and Future Directions

It has been an eye-opener in this module, both personally and professionally. I have not only obtained a clearer understanding of machine learning algorithms but also heard about the ethical features of AI technologies. I am going to continue advancing my knowledge in the area of machine learning in the future, particularly in such areas as deep learning and natural language processing, which will also broaden my skills.

I also shall practice the application of the skills to more sophisticated information and problems in life. When I move to a professional environment, I am hoping to use machine learning to address valuable problems and make sure that the considerations related to ethics guide my actions. The experience acquired in this module will definitely affect my further work with the projects and help me become a responsible developer and implementer of AI technologies.

Reflection on Machine Learning Module

1. Project Overview and Key Aspects (WHAT)

Throughout this module, I managed to complete two major projects: predicting prices of Airbnb listings in New York City and speaking about the ethical side of AI writers.

- **Airbnb Price Prediction:** The overall concept of the project was to predict the price of Airbnb houses relying on various features, including the room type, the location, and the availability. I used regression (Linear Regression, Ridge Regression and Random Forest Regression) and clustering (K-Means) to locate market segments.
- **AI Writers:** The second subject of the module was the consciousness of the potential threats and opportunities of AI writers. We talked about their applications in administration and creative writing, and the moral aspects of the development of biased or false materials.

The module was vital in enhancing my technical expertise in machine learning and advancing my understanding of the ethical aspects of AI. Some of the activities that I completed in the project are preprocessing of the dataset, feature extraction, model selection, and performance evaluation. The other factor that made me have a moderating perception of the impacts of AI in different fields was the ethical examination of AI authors that I took part in.

2. Emotional Response and Analysis (SO WHAT)

Initial Feelings

I became enthusiastic about the idea of applying machine learning algorithms to real-world data and feared my possible ability to accomplish complex tasks (feature engineering, model evaluation, and working with large datasets) when I first enrolled in the module. The course

required both theoretical understanding of algorithms and a practical application of the knowledge, which was challenging at first stage.

Team Collaboration

There was also a problem with group work. We had at one time misunderstood each other regarding the models to apply and the manner in which we were going to deal with some of the data, which initially made it tense. Nevertheless, these were good learning experiences. They taught me to express technical concepts effectively, positively solve conflicts, and work as a team towards a shared objective.

The collaborative work of the team in refining the models and making everything in the project evaluation process complete contributed to the success of the project. I was glad to feel an accomplishment when the model performance improvements were observed after applying hyperparameter optimisation and cross-validation.

AI Writer Analysis

The ethical implications of AI writers are also a mixed emotion. On the one hand, the opportunities of AI helping in creative work attracted my attention, and I also feared the possible dangers of AI-generated content, particularly in spheres that demand high precision, such as healthcare and law. I started viewing AI not as a substitute for human creativity but as an instrument that can complement it, which made me form a more consistent opinion of AI technologies.

3. Learning and Changed Actions (NOW WHAT)

Knowledge Development

Throughout the module, I have acquired a lot of knowledge on machine learning algorithms. First, I had a problem with the choice and use of the appropriate models on the data. Nonetheless, a combination of practical experience helped me to gain a better insight into when a linear model should be employed and when it is necessary to use non-linear models, such as the Random Forest, and to test the model efficiency with the help of the necessary metrics.

I also got to know the significance of preprocessing data, particularly the processing of missing data, as well as manipulating the variables in such a way that the models could optimise their performance. Such practical skills will play a significant role in future projects where data quality is the main priority.

Another dimension of the ethical analysis of AI writers that I acquired novel knowledge in was in the field. I did not think of the dangers of bias or hallucinations of AI-generated content before. The awareness of these problems has helped me be more aware of implementing AI in high-stakes settings without appropriate supervision.

Changes in Approach

Later on, I changed my strategy towards model choice. First, I was more concerned with theory, but as I started working with actual data, I began to understand that practicality and understandability were equally significant as accuracy. I started paying more attention to not only accurate models, but also transparent models, and this is why I preferred Random Forest to more complicated deep learning models in the case of the given project.

Another change in my opinion concerning AI technologies was the moral debate on AI writers. I altered my view of AI as a potential replacement for human labour to its use as a tool that has to be implemented with caution, especially in creative and high-stakes fields.

4. Evidence of Skills and Knowledge Development

Artefact 1: Airbnb Price Prediction

The practice of working with the Airbnb data using the Random Forest regression provided me with concrete evidence of my abilities in applying machine learning techniques. Both my competence in the field of data science and machine learning increased because of the effectiveness of the model in price forecasting and the efficient use of hyperparameter optimisation.

Artefact 2: Clustering Analysis Using K-Means

The clustering analysis using K-Means showed that I was able to subdivide the data and make useful conclusions. This was a highly significant step towards the actualisation of the manner in which the different sections of the market can be targeted in the context of maximising the price.

Artefact 3: Ethical Analysis of AI Writers

My ethical reflection on AI writers showed that I was able to critically evaluate the greater implications of emerging technologies on society. I also covered the problem of data bias, copyright, and the necessity, which becomes essential when implementing AI in the real world.

5. Conclusion and Future Directions

The module has been critical to my learning of machine learning algorithms, data preprocessing, and the ethical implications of AI. I am also better placed to tackle machine learning problems in future projects, and I have a clearer picture of how AI can be utilised ethically in other fields.

I would also like to continue learning more about advanced machine learning practices in the future and explore the issue of AI technologies and ethics. With the further development of AI, I will be

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dedicated to the idea that it should be applied in an ethical and responsible way to address the challenging real-world issues.

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